Warmup

Biochemistry PhD students

We have information on 915 biochemistry PhD students, with variables including:

- `art`: articles published by the student in last three years of Ph.D.
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program (a numeric score from 0 to 5)
- `ment`: articles published by their mentor in last three years

We are interested in modeling the number of articles published by the student in the last three years of their PhD, and so we fit a Poisson regression model:

```
m_art <- glm(art ~ phd + ment + kid5,
             family = poisson, data = articles)
summary(m_art)
```

```
Call:
glm(formula = art ~ phd + ment + kid5, family = poisson, data = articles)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.276493   0.087294   3.167  0.00154 **
phd          0.008071   0.026358   0.306  0.75944
ment         0.026537   0.002001  13.264  < 2e-16 ***
kid5        -0.099696   0.035248  -2.828  0.00468 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1817.4  on 914  degrees of freedom
Residual deviance: 1661.1  on 911  degrees of freedom
AIC: 3336.8

Number of Fisher Scoring iterations: 5
```

Questions

1. How would we interpret the coefficient on the number of articles published by the student's mentor?

2. What is the predicted number of articles for a student at a PhD program with a prestige score of 4, whose mentor published 30 articles in the last 3 years, and who has no children?

3. What assumptions is this model making? How would we assess those assumptions?